

Grupo G23.

Módulo 1. Recuperatorio. 9/10/24

$$1) \quad q_1 = 2q \quad \langle 0,0 \rangle \quad q_3 = -q \quad \langle a,0 \rangle \quad P \left(\frac{2}{3}a; \frac{2}{3}b \right) \\ q_2 = q \quad \langle 0,b \rangle \quad q_4 = 3q \quad \langle a,b \rangle; \quad q = 9 \cdot 10^{-9} \text{ C} \\ a = 0,15 \text{ m}; b = 0,24 \text{ m}$$

$$A) \quad W_{4P} = (V_P - V_4) \cdot q_4. \quad \text{Referencia: } V(\infty) = 0$$

$$V_P = V_{P1} + V_{P2} + V_{P3}$$

$$V_{P1} = \frac{kq_1}{r_{P1}}; \quad \vec{r}_{P1} = \vec{r}_P - \vec{r}_1 = \left\langle \frac{2}{3}a; \frac{2}{3}b \right\rangle - \langle 0,0 \rangle = \left\langle \frac{2}{3}a; \frac{2}{3}b \right\rangle$$

$$r_{P1} = \sqrt{\frac{4}{9}a^2 + \frac{4}{9}b^2} = \frac{2}{3} \sqrt{a^2 + b^2} = 0,189 \text{ m}$$

$$V_{P1} = \frac{9 \cdot 10^9 \cdot 2q}{\frac{2}{3} \sqrt{a^2 + b^2}} = \frac{9 \cdot 10^9 \cdot 2 \cdot 9 \cdot 10^{-9}}{\frac{2}{3} \sqrt{0,15^2 + 0,24^2}} = 857,14 \text{ V.}$$

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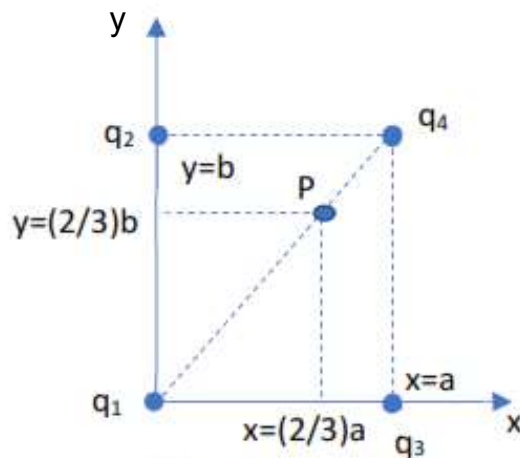


Figura 1

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$$V_{P2} = \frac{k q_3}{r_{P2}} ; \quad \vec{r}_{P2} = \vec{r}_P - \vec{r}_2 = \left\langle \frac{2}{3}a, \frac{2}{3}b \right\rangle - \langle 0, b \rangle$$

$$= \left\langle \frac{2}{3}a, -\frac{1}{3}b \right\rangle$$

$$r_{P2} = \sqrt{\frac{4}{9}a^2 + \frac{1}{9}b^2} = \frac{1}{3} \sqrt{4a^2 + b^2} = 0,128 \text{ m}$$

$$V_{P2} = \frac{9 \cdot 10^9 \cdot 9 \cdot 10^{-9}}{0,128} = 632,81 \text{ V}$$

$$V_{P3} = \frac{k q_3}{r_{P3}} ; \quad \vec{r}_{P3} = \vec{r}_P - \vec{r}_3 = \left\langle \frac{2}{3}a, \frac{2}{3}b \right\rangle - \langle a, 0 \rangle$$

$$\vec{r}_{P3} = \left\langle -\frac{1}{3}a, \frac{2}{3}b \right\rangle ; \quad r_{P3} = \sqrt{\frac{1}{9}a^2 + \frac{4}{9}b^2}$$

$$r_{P3} = \frac{1}{3} \sqrt{0,15^2 + 4 \cdot 0,24^2} = 0,168 \text{ m}$$

$$V_{P3} = \frac{9 \cdot 10^9 \cdot (-9 \cdot 10^{-9})}{0,168} = -482,14 \text{ V}$$

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$$V_P = 1007,81 \text{ V}$$

$$V_4 = V_{41} + V_{42} + V_{43}$$

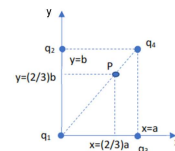
$$V_{41} = \frac{k q_1}{r_{41}} ; \quad \vec{r}_{41} = \vec{r}_4 - \vec{r}_1 = \langle a, b \rangle - \langle 0, 0 \rangle = \langle a, b \rangle$$

$$r_{41} = \sqrt{a^2 + b^2} = 0,283 \text{ m}$$

$$V_{41} = \frac{9 \cdot 10^9 \cdot 9 \cdot 10^{-9} \cdot 2}{0,283} = 572,43 \text{ V}$$

$$V_{42} = \frac{k q_2}{r_{42}} ; \quad r_{42} = a = 0,15 \text{ m}$$

$$V_{42} = \frac{9 \cdot 10^9 \cdot 9 \cdot 10^{-9}}{0,15} = 540 \text{ V}$$



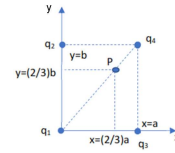
4

$$V_{43} = \frac{k q_3}{r_{43}} ; \quad r_{43} = b = 0,24 \text{ m}$$

$$V_{43} = \frac{9 \cdot 10^9 \cdot (-9 \cdot 10^{-9})}{0,24} = -337,5 \text{ V}$$

$$V_A = 774,93 \text{ V}$$

$$W_{AP} = (1007,81 - 774,93) \cdot 3 \cdot 9 \cdot 10^{-9} = 6,3 \cdot 10^{-6} \text{ J}$$



$$\text{B)} \quad U_{\text{sist}} = \frac{k q_1 q_2}{r_{12}} + \frac{k q_1 q_3}{r_{13}} + \frac{k q_1 q_4}{r_{14}} + \frac{k q_2 q_3}{r_{23}} + \frac{k q_2 q_4}{r_{24}} + \frac{k q_3 q_4}{r_{34}}$$

$$r_{12} = b = 0,24 \text{ m} ; \quad r_{14} = \sqrt{a^2 + b^2} = 0,283 \text{ m}$$

$$r_{13} = a = 0,15 \text{ m} ; \quad r_{23} = 0,283 \text{ m} ; \quad r_{24} = a = 0,15 \text{ m}$$

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$$r_{34} = b = 0,24 \text{ m}$$

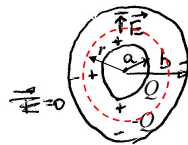
$$\therefore U_{\text{sist}} = k q^2 \left(\frac{2}{0,24} + \frac{(-2)}{0,15} + \frac{6}{0,283} + \frac{(-1)}{0,283} + \frac{3}{0,15} + \frac{(-3)}{0,24} \right)$$

$$U_{\text{sist}} = 9 \cdot 10^9 \cdot (9 \cdot 10^{-9})^2 \cdot (20,17) = 14,7 \mu\text{J}$$

P2) a)

$$a = 0,02 \text{ m}$$

$$b = 0,078 \text{ m}$$



$$E(r) = \begin{cases} 0 & 0 \leq r < a \\ \frac{kQ}{r^2} & a < r < b \\ 0 & r > b \end{cases}$$

$$V(r) = \begin{cases} kQ \left(\frac{1}{a} - \frac{1}{b} \right) & r < a \\ kQ \left(\frac{1}{r} - \frac{1}{b} \right) & a \leq r \leq b \\ 0 & r > b \end{cases}$$

Tomamos referencial
en el ∞ .
 $V(\infty) = 0$

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$$C_1 = \frac{Q}{\Delta V} = \frac{Q}{kQ \left(\frac{1}{a} - \frac{1}{b} \right)} = \frac{4\pi\epsilon_0 ab}{(b-a)} = 3 \text{ pF}$$

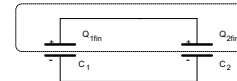
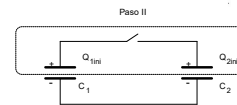
$$b) Q_{1f} = C_1 \cdot \epsilon = 48 \text{ pC}$$

$$c) C_1 = 3 \text{ pF}; C_2 = 5 \text{ pF}; C_{eq} = 8 \text{ pF}$$

$$V_f = \frac{Q_{1i} + Q_{2i}}{C_{eq}} = \frac{48}{8} = 6 \text{ V}$$

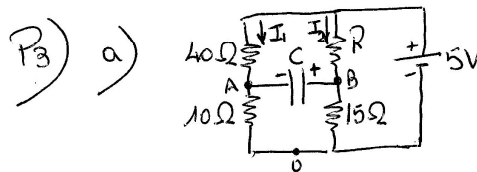
$$Q_{1f} = 18 \text{ pC}; Q_{2f} = 30 \text{ pC}$$

RECORDAR QUE:



$$Q_{1ini} + Q_{2ini} = Q_{1fin} + Q_{2fin}$$

$$Q_{neta, fin} = (C_1 + C_2) V_f = C_{eq} \cdot V_f$$



$$R = 10 \Omega$$

Por el capacitor no circula corriente.

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$$I_1 = \frac{5}{50} = 0,1 \text{ A}$$

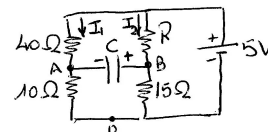
$$I_2 = \frac{5}{25} = 0,2 \text{ A}$$

$$V_{AO} = I_1 \cdot 10 = 1 \text{ V}$$

$$V_{BO} = I_2 \cdot 15 = 3 \text{ V}$$

$$V_{BO} - V_{AO} = V_{BA} = 2 \text{ V} \Rightarrow Q = V_{BA} \cdot C = 2 \text{ nC}$$

b) $R \cdot 10 = 40 \cdot 15$ (El producto de las resistencias cruzadas debe ser el mismo).
 $R = 60 \Omega$ No depende de la fem.



Punto b)

$$V_{AO} = V_{BO} \text{ para que } Q = 0$$

$$I_1 = \frac{E}{(40+10)}; V_{AO} = I_1 \cdot 10 = \frac{E \cdot 10}{(40+10)}$$

$$I_2 = \frac{E}{(R+15)}; V_{BO} = I_2 \cdot 15 = \frac{E \cdot 15}{(R+15)}$$

$$\therefore \frac{E \cdot 10}{(40+10)} = \frac{E \cdot 15}{(R+15)} \quad \text{cancelando } E \cdot 10 \cdot 15$$

$$\Rightarrow (R+15) \cdot 10 = (40+10) \cdot 15$$

$$R \cdot 10 = 40 \cdot 15$$

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P4) los tres cables rectos que pasan por el origen no generan $\vec{B}$ en el punto O.

$$\vec{B}_{OI} = \frac{\mu_0 I_2}{(2 \cdot R_2) \cdot 8} (-\hat{k}) = \frac{\mu_0 \cdot I_2}{16 \cdot R_2} (-\hat{k}) = \frac{4\pi \cdot 10^{-7} \cdot 6}{4 \cdot 16 \cdot 0,75 \cdot 10^{-2}} (-\hat{k}) = 62,8 \cdot 10^{-6} (-\hat{k}) \text{ T}$$

$$\vec{B}_{OII} = \frac{\mu_0 I_2}{(2R_1) \cdot 4} \hat{k} = \frac{\mu_0 \cdot I_2}{8R_1} (\hat{k}) = \frac{4\pi \cdot 10^{-7} \cdot 6}{2 \cdot 8 \cdot 0,50 \cdot 10^{-2}} (\hat{k}) = 188,5 \cdot 10^{-6} (\hat{k}) \text{ T}$$

$$\vec{B}_{OIII} = \frac{\mu_0 I_1}{2\pi d} (\hat{k}) = \frac{4\pi \cdot 10^{-7} \cdot 12}{2\pi \cdot 0,25 \cdot 10^{-2}} (\hat{k}) = 960 \cdot 10^{-6} (\hat{k}) \text{ T}$$

$$\vec{B}_{OT} = \vec{B}_{OI} + \vec{B}_{OII} + \vec{B}_{OIII} = (-62,8 + 188,5 + 960) \hat{k} [\mu\text{T}]$$

$$\vec{B}_{OT} = 1085,7 (\hat{k}) [\mu\text{T}] \approx 1,1 (\hat{k}) \text{ mT}$$

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P4) los tres cables rectos que pasan por el origen no generan $\vec{B}$ en el punto O.

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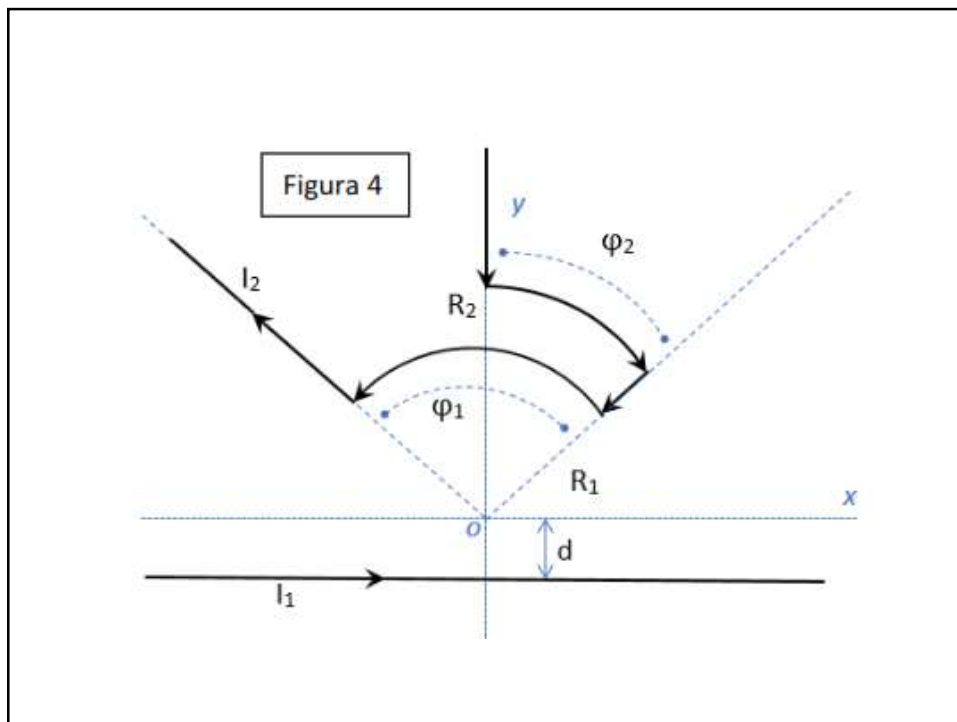
$$\vec{B}_{OII} = \frac{\mu_0 I_2}{(2R_1) \cdot 4} \hat{k} = \frac{\mu_0 \cdot I_2}{8R_1} (\hat{k}) = \frac{4\pi \cdot 10^{-7} \cdot 6}{2 \cdot 8 \cdot 0,50 \cdot 10^{-2}} (\hat{k}) = 188,5 \cdot 10^{-6} (\hat{k}) \text{ T}$$

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$$\vec{B}_{OT} = \vec{B}_{OI} + \vec{B}_{OII} + \vec{B}_{OIII} = (-62,8 + 188,5 + 960) \hat{k} [\mu\text{T}]$$

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P5)

a)

$$B(r) = \frac{\mu_0 i_1(t)}{2\pi r}$$

$$d\Phi = \frac{\mu_0 i_1(t)}{2\pi r} b \cdot dr$$

$$\Phi(t) = \frac{\mu_0 i_1(t)}{2\pi} b \int_c^{c+a} \frac{dr}{r}$$

$$\Phi(t) = \frac{\mu_0 i_1(t) \cdot b}{2\pi} \ln\left(\frac{c+a}{c}\right)$$

$$\Phi(t) = \frac{4\pi \cdot 10^{-7} \cdot (5+5t) \cdot 0,45}{2\pi} \ln\left(\frac{25}{10}\right) = 4,5 \cdot 10^{-7} (1+t) \ln(2,5)$$

$$\Phi(t) = 4,12 \cdot 10^{-7} (1+t) [Wb]$$

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$$b) \quad \mathcal{E}_i = -N \frac{d\Phi}{dt} = -N \cdot 4,12 \cdot 10^{-7} \left[\frac{Wb}{s} \right]$$

$$|\mathcal{E}_i| = N \cdot 4,12 \cdot 10^{-7} = 0,206 \text{ mV} \Rightarrow N = 500 \text{ vueltas.}$$

$$c) \quad \text{Si } R = 20 \Omega$$

$$|\mathcal{E}_i| = 0,206 \text{ mV} \quad i_{\text{ind}} = \frac{0,206 \cdot 10^{-3}}{20} = 10,3 \mu\text{A}$$

Como el Φ ↑ con el tiempo, la corriente inducida se opone al cambio

$$i_{\text{ind}} \downarrow \begin{array}{|c|} \hline \otimes \vec{B} \\ \hline \odot \vec{B}_{\text{ind}} \\ \hline \end{array} \frac{1+}{1-} \mathcal{E}_i$$

$$d) \quad \mathcal{E}_i = -M \frac{di_1}{dt} = -M \cdot \frac{d}{dt}(5+5t) = -M \cdot 5 = -0,206 \cdot 10^{-3} \text{ V}$$

$$M = 41,2 \mu\text{H.}$$